



△ **Faceted 7.79-carat** indicolite tourmaline, side view

Tourmaline

Tourmaline refers to a family of borosilicate minerals of variable composition, but all with the same basic crystal structure. There are more than 30 mineral species in the tourmaline group, including elbaite, dravite, and schorl. However, while mineral names are based on chemistry, gemstone names are based on colour and take no notice of tourmaline species. These include indicolite (blue), achroite (colourless), and rubellite (pink or red). The crystals generally form pencil-like prisms, with a rounded-triangular cross-section, and, unlike the rocks in which they often form, tourmaline minerals are resistant to weathering. As a result, they tend to accumulate in gravel deposits; the origin of the name is the Singhalese word *turamali* – “gem pebbles”.

Variety of colours

There is no simple correlation between chemical composition and colour. Most gemstone tourmaline material comes from the species elbaite, which is usually green, although it can occur in many other colours. Emerald green is fairly rare and thus valuable; until the 18th century, it was often confused with emerald. The most dramatic tourmalines are the colour-zoned gems called “watermelon” tourmaline: when sliced across the crystal, this variety shows a red or pink centre surrounded by a rim of green. The deepest colour is always seen when looking down the length of the crystal, so it is important to position rough material correctly when cutting gems.

Key pieces



Gold and tourmaline necklace | This 18-karat gold necklace is set with pear-shaped tourmalines weighing a total of approximately 100 carats, accented with brilliant-cut diamonds.



Tiffany brooch | Created for Tiffany & Co. by Jean Schlumberger, this 18-karat textured gold salamander brooch is set with rectangular-cut green tourmalines, diamond feet, and turquoise cabochon eyes.

Specification

Chemical name Complex boron silicate | **Formula** $\text{Na}(\text{Li}_{1.5}\text{Al}_{1.5})\text{Al}_6(\text{BO}_3)_3[\text{Si}_6\text{O}_{18}](\text{OH})_3(\text{OH})$ (elbaite) | **Colours** Various
Structure Trigonal | **Hardness** 7–7.5 | **SG** 2.8–3.3
RI 1.61–1.67 | **Lustre** Vitreous | **Streak** White



Emerald



Mixed



Marquise



Pendalogue



Step



Cabochon



Locations

1 USA **2** Brazil **3** Czech Republic **4** Italy **5** Nigeria
6 Namibia **7** South-eastern Africa **8** Madagascar
9 Afghanistan **10** Pakistan **11** Sri Lanka **12** Australia



Cartier earrings | These fanciful orchid earrings, set in 18-karat gold, are studded with faceted pink tourmalines, pink sapphires, rhodolite garnets, and 24 diamonds. A single briolette-cut rose quartz drop completes the earrings.